

Title

University of Glasgow



University
of Glasgow

Joseph Henry Briggs

Day Month Year

Contents

1	Balanced Homodyne for LIGO A+	4
1.1	What is BHD	4
1.2	BHD versus DC readout for A+	9
1.3	Local Oscillator Path Length Stability Requirement	15
1.3.1	Local Oscillator Path	16
1.3.2	The Linear Response Function Between Differential Strain and Power at the AS Port	19
1.3.3	Converting the Sensitivity Curve for A+ into a Longitudinal Motion Requirement	21
1.3.4	Summary of LO path length Stability	26
1.4	Conclusion	26
2	Design of the HAM6 layout	27
2.1	The Constraints and Requirements of the Layout of HAM6	28
2.2	Chosen Layout	34
2.3	WS visulations for Analysing the Robustness of the HAM6 Layout to Mode-Mismatch	40
2.4	Conclusion about BHD for A+	48

A	Quantum Optics	49
A.1	Definitions	49
A.2	single photon state and a beam splitter	50
A.3	2 Single photon states entering the beam splitter	51
A.4	coherent state and a beam splitter	52
A.5	2 coherent states and a beam splitter	52
A.6	squeezed state and a beam splitter	55
A.7	coherent state, squeezed state and a beam splitter	56

Chapter 1

Balanced Homodyne for LIGO A+

This chapter explores the advantages, requirements and practical design considerations for the implementation of balanced homodyne detection (BHD) for A+. Section 1.1 describes the fundamentals of BHD. Section 1.2 compares BHD to DC readout, which is the current readout scheme employed by aLIGO. Section 1.3 derives the phase noise requirement for the local oscillator (LO).

1.1 What is BHD

BHD is a method of measuring information encoded by a light field, and it forms a key part of many quantum optics experiments. By combining a signal beam with a much stronger LO beam, phase modulations in the signal beam become amplitude modulations in the combined LO-signal beam. BHD can be configured to measure any quadrature of the signal beam, which is vital for inferring the quantum state of a system and is required for more complex topologies of interferometer.

All readout schemes for gravitational wave detectors rely on a LO field of some kind. The first generation of detectors used heterodyne readout. Heterodyne readout involves a modulation-demodulation scheme whereby the signal is mixed with a RF local oscillator to a frequency where technical noise is no longer a concern and then is demodulated at the same modulation frequency to recover the original signal. This method comes with the draw back that the quantum shot noise is enhanced by the modulation-demodulation process.

The LO in a BHD readout scheme is a light field with the same optical frequency as the signal beam, making BHD a special case of heterodyne detection. The LO beam has a higher power than the signal beam and acts as a stable reference frequency. Ideally the LO would have no phase noise, however in reality there are fluctuations in the phase between the LO and the signal beam. The interference of the LO and signal beam causes this phase noise to become an amplitude modulation in the combined LO-signal beam. This phase noise must be kept below the size of the phase fluctuations in signal beam that are to be measured, which in A+ are determined by the shot noise of the light within the arm cavities. Unlike Heterodyne readout, BHD does not necessarily cause an increase in the noise floor of the measurement.

BHD is implemented by injecting the LO and signal beams into different input ports of a balanced beam splitter and placing photo-diodes in the remaining output ports. The information encoded in an arbitrary quadrature of the signal beam can be selected by correctly setting the relative phase between the LO and signal beams. The relative phase between the LO and signal beam is also known as the *homodyne angle*. Subtracting the two photo-currents results in a measurement which

is proportional to the selected quadrature of the signal beam. A simple BHD layout is shown in Figure 1.1.

The following few paragraphs of maths will likely be re written as I come to understand quantum optics more deeply. In particular, the equation for the BHD signal will be reworked, and an accurate description of the noise will be written. Appendix A contains a work in progress of this. I was not satisfied with the quantum description in [1] as it seemed a bit hand wavy. For instance, Section II of [1] describes splitting the field into a classical and quantum part and explains it with ‘linearisation’. In reality, a coherent state is still a quantum one (otherwise it would have no zero point fluctuations!), so I find splitting it into a ‘classical’ and ‘quantum’ to be misleading. I think the same results can be obtained with the exact same level of mathematics with a fully quantum explanation. It is true that coherent states have lots of similarities to the same way as a classical beam, but beamsplitters will violate basic quantum mechanics if classical assumptions are used with them. Instead, I will refer to Gerry and Knight.

Due to the balanced nature of the beam splitter, the classical intensity noise of the local oscillator is not present in the BHD signal. These properties of BHD make it ideal for measuring quantum states of light. For example, BHD can be used to detect the level of squeezing in quadrature of the signal beam by setting the homodyne angle to the quadratures angle.

BHD is suited to inferring the quantum state of two-photon phenomena, such as squeezing, as two photo-diodes are required to detect two photons simultaneously. A

two-photon process can be described in terms of the amplitude and phase quadrature operators, denoted as $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ respectively (see Caves[2]). These operators are related to the annihilation and creation operators of the electromagnetic field, $\hat{a}^\dagger(t)$ and $\hat{a}(t)$, by

$$\hat{\mathbf{X}}(t) = \frac{1}{\sqrt{2}} (\hat{a}(t) + \hat{a}^\dagger(t)) \quad (1.1)$$

$$\hat{\mathbf{Y}}(t) = \frac{1}{\sqrt{2}} (\hat{a}(t) - \hat{a}^\dagger(t)). \quad (1.2)$$

For a two-photon process, the general quadrature operator is

$$\hat{\mathbf{X}}_\phi = \hat{\mathbf{X}} \cos \phi + \hat{\mathbf{Y}} \sin \phi. \quad (1.3)$$

Consider a coherent state, $|\alpha_{\text{ref}}\rangle$ being used as a reference beam, and an arbitrary state in the signal beam. They are mixed on a beam splitter and detected on two photodiodes, as shown in figure 1.1. The two photo-diodes detect

$$\hat{\mathbf{N}}_{\text{c}} = \hat{c}^\dagger \hat{c} = \left(\sqrt{R} \hat{a}^\dagger + \sqrt{T} \hat{b}^\dagger \right) \left(\sqrt{R} \hat{a} + \sqrt{T} \hat{b} \right) \quad (1.4)$$

$$= R \hat{a}^\dagger \hat{a} + T \hat{b}^\dagger \hat{b} + \sqrt{RT} (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}) \quad (1.5)$$

$$\hat{\mathbf{N}}_{\text{d}} = \hat{d}^\dagger \hat{d} = T \hat{a}^\dagger \hat{a} + R \hat{b}^\dagger \hat{b} - \sqrt{RT} (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}). \quad (1.6)$$

For balanced homodyne detection, $R = T = 1/2$ so

$$\hat{\mathbf{N}}_{\text{c}} - \hat{\mathbf{N}}_{\text{d}} = \hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a} \approx \alpha_{\text{ref}}^* \hat{b} + \alpha_{\text{ref}} \hat{b}^\dagger. \quad (1.7)$$

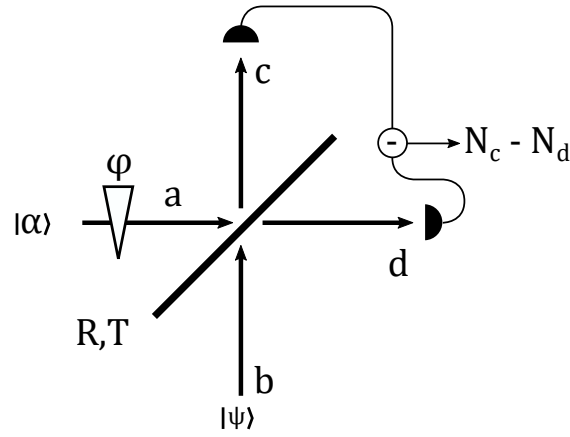


Figure 1.1: A coherent state $|\alpha\rangle$, enters port a of a beam splitter. A signal $|\Psi\rangle$ enters port b . These two beams interfere, resulting in photons being detected at the c and d ports. The two photo-currents are subtracted, and the signal that is produced is proportional to an arbitrary quadrature of the state $|\Psi\rangle$. The quadrature that is measured is determined by the homodyne angle, ϕ . This is expressed mathematically by equation 1.11.

The phase between the signal and the reference is θ , so the reference beam eigenvalue may be expressed as

$$\alpha_{\text{ref}} = |\alpha_{\text{ref}}| \exp^{i\phi}. \quad (1.8)$$

The creation and annihilation operators, $\hat{b}$ and $\hat{b}^\dagger$ can be written in terms of quadrature operators

$$\hat{b} = \frac{1}{\sqrt{2}} (\hat{\mathbf{X}} + i\hat{\mathbf{Y}}), \quad (1.9)$$

$$\hat{b}^\dagger = \frac{1}{\sqrt{2}} (\hat{\mathbf{X}} - i\hat{\mathbf{Y}}), \quad (1.10)$$

allowing for equation (1.7) to be written as

$$\hat{\mathbf{N}}_{\text{c}} - \hat{\mathbf{N}}_{\text{d}} \approx \left| \sqrt{2}\alpha_{\text{ref}} \right| (\hat{\mathbf{X}} \cos \phi + \hat{\mathbf{Y}} \sin \phi). \quad (1.11)$$

This shows us that the photo-current will be proportional to the general quadrature operator, hence enabling us to infer the quantum state of the system. This useful thesis may contain other perspectives.

1.2 BHD versus DC readout for A+

There are some advantages for BHD over DC readout, although for A+ a quantifiable difference is hard to make. The main attraction for implementing BHD is the ability to reduce the dark fringe offset that is required to produce the LO for DC readout.

This offset gives rise to most of the limitations associated with DC readout, such as the coupling between the signal recycling cavity length and the DARM signal. BHD also provides the ability to select which quadrature of the signal beam is measured, which is a requirement for novel interferometer configurations.

The current detection scheme is known as DC readout. A differential offset in the arm lengths produces the LO in DC readout. The LO beats with a signal created by a gravitational wave in order to produce intensity modulations at the AS port. The size of the LO can be controlled with the DC offset, and the homodyne angle can be controlled with a differential arm power. This method of readout has some key positive outcomes, such as there being no phase noise between the LO and the signal, and the LO and signal being perfectly aligned and mode matched. The problems with this readout technique are associated with requiring the DC offset as this light gives rise to noise couplings.

For BHD to be worthwhile it must be at least as good as DC readout and so BHD should be considered in relation to the perks of DC readout. Unlike DC readout, BHD may introduce phase noise between the LO and signal. Therefore the relative stability between the LO and signal beams has to be considered. Control over the alignment and mode matching between the LO and the signal is also important. LO light that is scattered into higher order modes, by misalignment or mode mismatch, may backscatter from the OMC into the interferometer and could result in noise in the DARM signal. The implementation of BHD will only be successful if it provides a stable enough LO beam; sufficient mode-matching between the LO and signal beam and active alignment of the LO and signal beams.

BHD has some practical benefits over DC readout¹ as well as some interesting prospects in relation to the quantum noise limit of the detector. The practical benefits arise due to being able to eliminate the DC light required for DC readout to work. The primary benefit is in the reduction in SRC length noise coupling to the gravitational wave signal. Another benefit is being able to reduce the scattering from the septum window between HAM5 and HAM6. Quantum noise shaping is, along with other interferometer enhancements, related to the complete free choice of homodyne angle that a BHD readout scheme permits.

One of the practical outcomes of implementing BHD is the potential to reduce SRC length noise coupling. Noise on the length of the SRC can couple to the gravitational wave signal via two routes. The dominant coupling will arise due to the power entering the SRC, which is proportional to the DC offset. The other coupling arises due to the optical spring between the SRC and the arm cavity length.

The dominant coupling due to the power entering the SRC can be intuitively understood by considering a Fabry-Perot cavity. For small changes in cavity length, the light transmitted by a cavity that is on resonance or anti-resonance will be modulated by a term proportional to the square of the length change. This signal

¹One ‘advantage’ that is often talked about in relation to BHD is based on the electronic noise of the readout photo-diodes. This is not real advantage. As the BHD signal is obtained through subtracting the signals from two photodiodes, it is possible to obtain a very low electronic noise floor when designing the transimpedance amplifier by subtracting the currents produced by the photodiodes rather than having two separate transimpedance amplifiers and subtracting the signals made by these later. This is not really an advantage, as low enough electronic noise can be obtained using other construction techniques that allow for small DC signals while still having signal shaping that gives a large enough SNR in the detection band. Having separate transimpedance amplifiers for each photodiode also has useful diagnostic advantages, such as being able to see the sum of the two photo-diode signals. Therefore, direct subtraction of the photo-diode signals should not be considered an advantage of BHD over DC readout.

due to the length of the SRC is noise. The amplitude of this noise is linearly dependent on the amount of light entering the cavity which is a result of the DC offset. By decreasing the DC offset, this noise is decreased. Unlike the gravitational wave signal, as this noise is in the amplitude quadrature of the signal beam it will not be amplified by the LO.

The second route SRC length noise may couple to the the signal is via the optical springs between the SRC and arm cavity. Due to this optical spring effect, a change in length of the SRC will cause the arm cavity length to change as well[3]. This effect is much smaller than the first effect so the estimate for the reduction of SRC length coupling will only considering the first effect.

SRC length noise is currently too high to meet the A+ requirements. To estimate the reduction in SRC length noise that BHD may provide, the amount of DC offset that is currently used should be compared to the amount that is proposed for BHD. At the AS port there is 23 mW of light due to the DC offset and 1 mW of residual light (light that cannot be removed). The AS power for A+ is between 1 and 5 mW. Therefore, a conservative estimate for the factor that the SRC length noise coupling is reduced by is four.

Strictly speaking, these are not properties of BHD readout as scatter from the septum window and the SRC length noise could be reduced by other means. For instance, the septum window could be removed and suspensions with additional stages could be used. However, these solutions are not trivial to realise and they are no more practical than implementing BHD. The septum window is a key part in being able to carry out commissioning tasks within HAM6. The number of times

HAM5 and the other connected chambers are vented should be kept low as each time this is done gas and water vapour enters the system (this contributes to the noise floor of the interferometer). This residual gas and vapour will not be removed as this is a large engineering challenge. The design and install of three new suspensions is also a difficult task, which ultimately gives less benefit compared to a BHD readout scheme.

[4][5],[6][7][8][9]. Different gravitational wave sources carry their information in different frequency bands and the detector may be optimised to reflect this. For instance, BNS signals continue to higher frequencies than BBH signals (due to them being smaller) so it may be of interest to optimise the detector around these frequencies to improve our understanding of the properties of BNS. To do this, the detectors noise would need shaping which is a task that BHD is vital for.

The current generation of gravitational wave detectors are going to be limited by a combination of radiation pressure noise and shot noise in their detection bands. These two noise sources are intrinsically linked by the Heisenberg uncertainty principle and can both be understood as arising due to the number of photons used to probe the position of the end test masses. This limit in the noise floor is known as the standard quantum limit. To circumvent this limit the homodyne angle must be free to choose; a functionality that is provided by BHD. In principle, DC readout gives the ability to set the homodyne angle although in practice it is much less flexible than a BHD configuration.

A squeezed light field can be used to surpass the standard quantum limit and to detect these squeezed states the homodyne angle must be correctly set. The

exact way in which the quantum noise is changed is controlled by the squeezing angle. In practice, the squeezing angle is set to be at the same angle as the readout quadrature, rather than the preferred option of setting the readout quadrature to the squeezing angle. The effect that controlling the squeezing angle has on the quantum noise floor has been demonstrated at LLO [5]. However, this experiment used a diagnostic BHD set-up which was not intended for providing the low noise required to make gravitational wave detections.

More advanced upgrades, e.g. see [8], could reduce quantum noise substantially, and they rely on the implementation of BHD. These changes to the interferometer are non-trivial, and would require significant redesign of many parts of the layout, so the implementation of BHD for A+ does not necessarily correspond to a suitable BHD system for one of these more advanced interferometer configurations. However, BHD for A+ would provide experience with implementing BHD on a large scale interferometer, which would help when considering how to implement these more novel interferometer configurations.

In summary, BHD has a couple of key advantages over DC readout. BHD allows for the reduction in SRCL noise coupling and, in conjunction with other upgrades, the ability to shape the quantum noise floor of the interferometer. In DC readout, the SCRL length coupling comes from the DC offset required to provide an LO beam for the signal generated by a differential modulation in arm length to beat against. This DC offset is not required for BHD. Most interferometer configurations that beat the SQL assume that the homodyne angle can be freely set. The drawbacks of BHD when compared DC readout are all engineering challenges; nothing fundamentally

favours DC readout over BHD.

1.3 Local Oscillator Path Length Stability Requirement

This section will go through the steps required to derive the phase noise requirement for the beam path of the A+ LO. A semi-classical approach has been used, and the end result is the same as if a full quantum mechanical treatment was used. In fact, the quantum mechanics of this problem is included in the A+ sensitivity curve and the expected power in the signal beam. The phase noise of the local oscillator must be kept low enough so that it does not result in intensity fluctuations greater than those due to the shot noise, which can be squeezed, of the light in the signal beam. This will be made more ‘bomb proof’.

Phase noise arises due to phase fluctuations in the LO beam being indistinguishable is phase fluctuations in the AS beam. A gravitational wave produces a phase modulated signal in the AS beam, and the LO field phase is modulated by the motion of the mirrors in the LO path. When these beams are combined, it is the phase difference between the two fields that produces a signal, so the phase noise of the LO path must be sufficiently low that the signal that is generated by the interference of the LO and AS beams corresponds to the phase modulation of the AS beam.

The relationship between the differential phase in the arms and the power in the signal beam was used to find the power noise for the beam exiting the SRM in A+. The linear response function due to a differential phase change in the arms was

computed with FINESSE. This was used to deconvolve the strain sensitivity curve to produce a power noise curve for the light exiting the AS port of the interferometer. Since the interferometer is linear, noise is transformed linearly. This means that the quantum nature of the noise is irrelevant to the calculation.

This light noise curve was then used to set the maximum allowable intensity fluctuations due to phase noise between the local oscillator and the signal beams. This phase noise can then be converted into a distance, and this produces a stability requirement for the steering optics used to combine the LO and signal beams.

The following Sections will go through the aforementioned steps. First the local oscillator path will be described. Then a linear response function between a difference strain and signal at the AS port will be obtained via FINESSE. Next, The sensitivity curve for A+ will be expressed in units of $\frac{W}{\sqrt{Hz}}$. This will be converted into a longitudinal motion requirement. Finally, there is a short discussion about the proposed optics that fulfil this requirement.

1.3.1 Local Oscillator Path

The beam path for the LO and AS beams up to the point of recombination will be described in this section. The local oscillator will be picked off from the power recycling cavity, and guided by one steering mirror, BHDM1, to the BHD combining beam splitter, BHDBS2. The signal beam will exit the OFI and be steered by one mirror, OM0, onto BHDBS2. These two mirrors will be used to align the beams so they overlap on the BHDBS2. The angle of BHDBS2 will be used to make the beams overlap in the far field.

The local oscillator could have been derived from a separate laser but, to acquire laser light with low frequency and intensity noise, the LO will be picked off from a point within the interferometer. Two options for the local oscillator options were considered: a beam picked off from the main beam splitters AR coating (BSAR), or the beam transmitted through PR2 [10] (POP). The determining factor in choice of pick off point is the effect of backscatter. The POP port was chosen as it is far superior to the BSAR choice at all frequencies and scatter phases. This is because backscattered light from the BSAR enters only one arm, thus making it a differential effect. Any backscatter that enters the POP port is common-mode, and therefore DARM is far less sensitive to this source of backscatter.

The layout for the optics which combine the the LO and AS beam is shown in Figure 1.2. The beam transmitted by PR2 is used for control of the interferometer so BHDBS1 is needed to pick off some of this light. The LO beam is steered towards the combining beam splitter BHDBS2 with BHDM1. Two lenses (**need names for these**) are used for mode matching the LO beam to the AS beam. The AS beam is transmitted through the SRM and OFI, and then it is steered towards BHDBS2 with OM0.

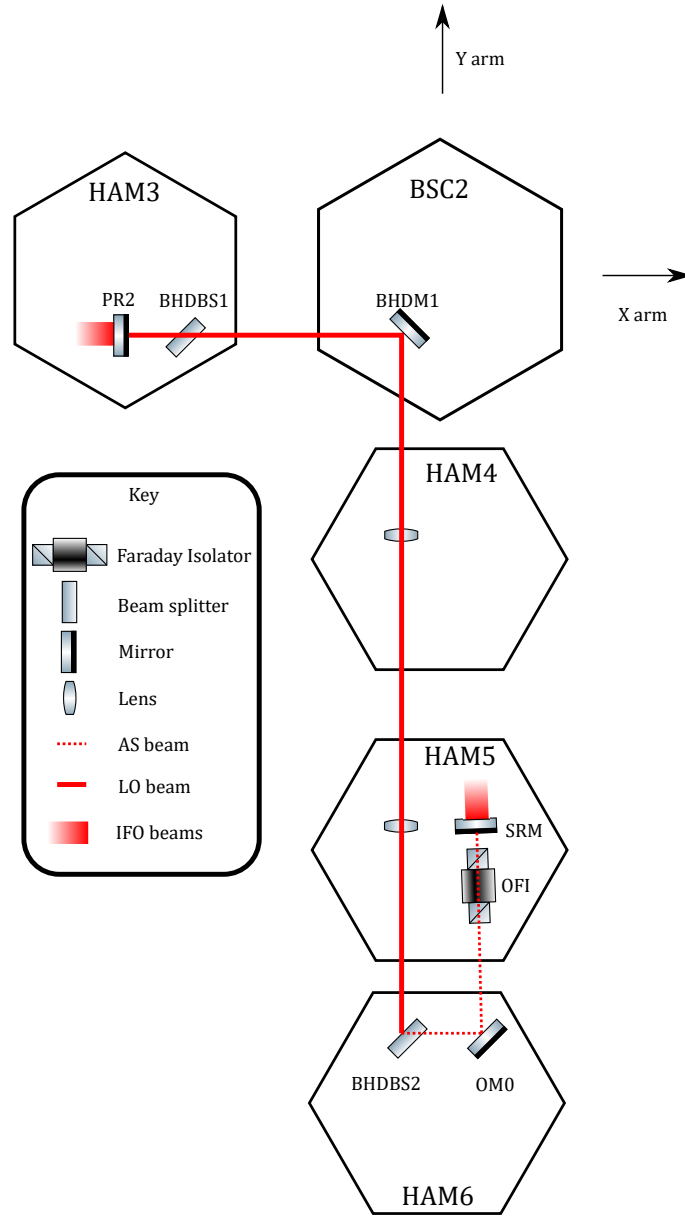


Figure 1.2: Sketch of the LO and AS beam paths. The HAM tables are shown by the hexagons. Most of the interferometer has not been shown for clarity. The main interferometer beam is indicated by a thick, gradated, red line. The thinner red line indicates the LO beam, and the signal beam is shown with a dashed red line.

1.3.2 The Linear Response Function Between Differential Strain and Power at the AS Port

The linear response function of the A+ detectors was obtained using FINESSE. Since the core optics of the interferometer will not be changing in a way relevant to this calculation, the LLO model[11] was used to calculate the response between differential motion of the ETMs and the power transmitted through the OMC. The effect of the light being transmitted through the OMC has no effect on the linear response function at the frequencies contained in the detection band, and so the OMC has no effect. This means no major modifications to the FINESSE model were needed i.e. all locking signals could be left alone.

FINESSE allows one to measure linear response functions by applying a ‘swept sine’ signal to a component and demodulating the light at the frequency of the applied signal. To apply a differential arm signal with FINESSE, `fsig` commands were used to apply two signals in antiphase with each other to the ETMs,

```
fsig sig1 ETMXHR 1 0
fsig sig2 ETMYHR 1 180.
```

The frequency of this modulation was swept and the signal produced at the AS port was measured by demodulating the light at the modulation frequency. The commands

```
xaxis sig1 f log 5 5000 2999
put P_DC_OMC_probe f1 $x1
```

do this.

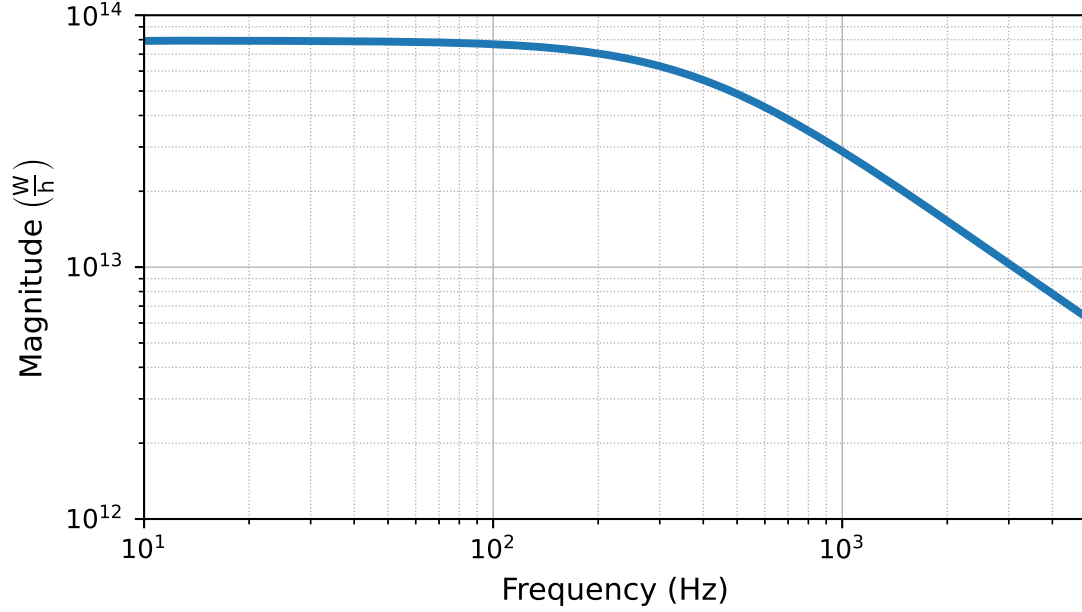


Figure 1.3: The linear response function between strain and light power, expressed in units of Light Power/gravitational wave strain, at the AS port was obtained from a FINESSE simulation of the LLO detector[11]. The corner frequency at ≈ 300 Hz comes from the pole of the coupled SRC-arm cavity.

For this calculation, the linear response function measured in FINESSE needs to be converted into different units. To get the linear response function between arm motion and AS power into units of $\frac{\text{Power}}{\text{Strain}}$, or $\frac{\text{W}}{\text{h}}$, several factors of 2 are needed. Since the input modulation is doubled by moving the mirrors differentially, the response needs dividing by 2. To account for the ETMs retro-reflecting the light, the response needs dividing by 2 again. To convert this from phase units to strain units, a factor of $\frac{\lambda}{2\pi L_{\text{arm}}}$ is needed. Therefore, the output of the the FINESSE simulation needs to be multiplied by $\frac{1}{2} \frac{1}{2} \frac{2\pi L_{\text{arm}}}{\lambda}$ to be in units of $\frac{\text{W}}{\text{h}}$. The linear response function between a differential arm strain and the power at the AS port is shown in Figure 1.3.

1.3.3 Converting the Sensitivity Curve for A+ into a Longitudinal Motion Requirement

The sensitivity curve will be shown in an earlier chapter. To convert the A+ sensitivity curve into a path length stability requirement for the local oscillator, a few steps are required. First, the sensitivity curve must be expressed in units of $\frac{W}{\sqrt{\text{Hz}}}$. Second, the intensity noise of the AS beam needs to be converted into a phase noise. In order for the phase noise of the local oscillator to not limit the performance of the gravitational wave detector, the phase signal in the AS beam must be greater than the phase noise of the LO beam. Finally, a factor of 10 safety margin is required. The HRTS requirement is then obtained by including the number of reflective optics in the LO path and signal path.

The A+ sensitivity[12] curve was multiplied by the function shown in 1.3 to convert it from units of $\frac{h}{\sqrt{\text{Hz}}}$ to $\frac{W}{\sqrt{\text{Hz}}}$. This is shown in figure 1.4. At frequencies above 1 kHz the requirement is flat as this is the region dominated by shot noise. At lower frequencies the requirement is relaxed as the sensitivity curve becomes dominated by other noise sources.

Next, a change in power measured in the combined LO/signal beam due to a phase modulation, i.e. path length modulation, in the LO or signal path needs calculating. When the LO and AS beams interfere at the BHSBS2 the exiting beam have power P_c and P_d . These are given by

$$P_c = \left| \frac{1}{\sqrt{2}} E_s e^{i\omega t} + \frac{1}{\sqrt{2}} E_{\text{LO}} e^{i\omega t + \phi} \right|^2 = \frac{|E_s|^2}{2} + \frac{|E_{\text{LO}}|^2}{2} + |E_s E_{\text{LO}}^\dagger| \cos(\phi) \quad (1.12)$$

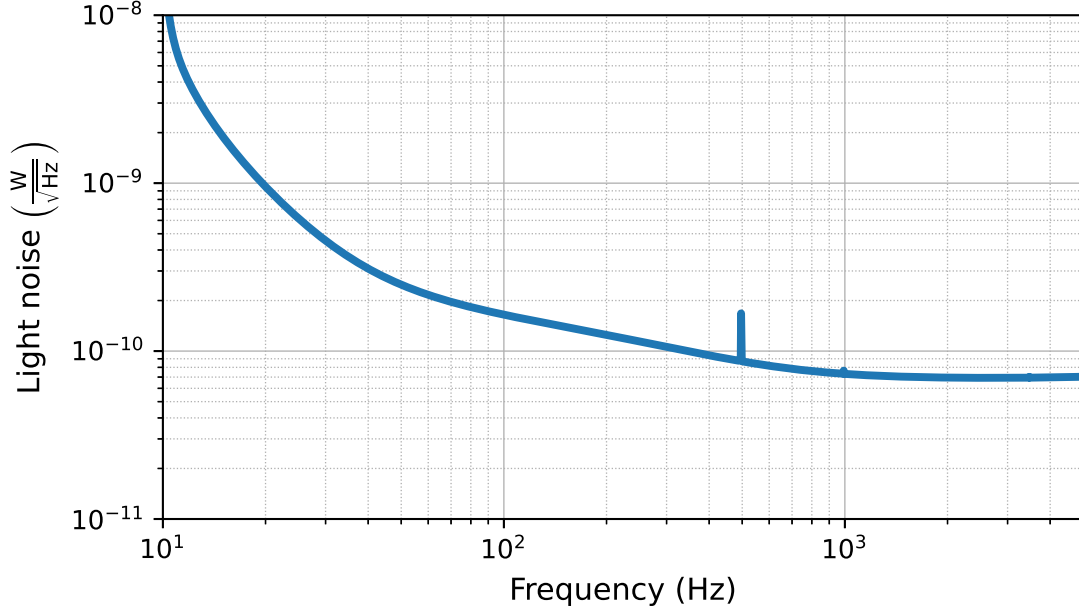


Figure 1.4: The intensity noise of the light at the AS port for the A+ detector. This was obtained by deconvolving the detectors sensitivity curve[12] with the function shown in figure 1.3. At frequencies above 1 kHz the noise is flat as it is dominated by shot noise. At lower frequencies the noise is increased relative to shot noise due to radiation pressure, coating thermal noise and other noise sources. The spike in the spectrum at 500Hz represents the presence of suspension thermal noise (?).

and

$$P_d = \left| \frac{1}{\sqrt{2}} E_s e^{i\omega t} - \frac{1}{\sqrt{2}} E_{LO} e^{i\omega t + \phi} \right|^2 = \frac{|E_s|^2}{2} + \frac{|E_{LO}|^2}{2} - |E_s E_{LO}^\dagger| \cos(\phi), \quad (1.13)$$

where E_s and E_{LO} are the (complex) field amplitude of the signal and local oscillator beams; ω is the laser frequency; t is time and ϕ is the phase difference between the local oscillator and the signal beam, also known as the homodyne angle. The

difference in these signal gives BHD signal

$$P = 2|E_s E_{\text{LO}}^\dagger| \cos(\phi). \quad (1.14)$$

Differentiating with respect to ϕ gives the relationship between a change in power and a change in phase between the two beams,

$$dP = -2|E_s E_{\text{LO}}^\dagger| \sin(\phi) d\phi. \quad (1.15)$$

and since the homodyne angle is 90° this becomes

$$dP = -2|E_s E_{\text{LO}}^\dagger| d\phi. \quad (1.16)$$

To convert ϕ to distance, use

$$\frac{\lambda}{2\pi} d\phi = dx. \quad (1.17)$$

The light noise curve shown in figure 1.4 should be multiplied by $\frac{\lambda}{2\pi} \frac{1}{2|E_s E_{\text{sig}}^\dagger|}$ to obtain the the noise floor of the interferometer in units of LO path length noise.

To determine the longitudinal motion requirement for the optics that combine the LO and signal beam the number of optics and a factor of ten safety margin needs to be included since there are many uncorrelated noise sources in LIGO. The requirement assumes, based on the current design, that there will be two mirrors at 45° which are used to steer the AS and LO beams. As the optics reflected at 45° , the path length change due to the optics motion is multiplied by $\frac{2}{\sqrt{2}} = \sqrt{2}$. The beams are combined on a beam splitter, and the motion of this beam splitter needs to be counted twice. Figure 1.5 illustrates this.

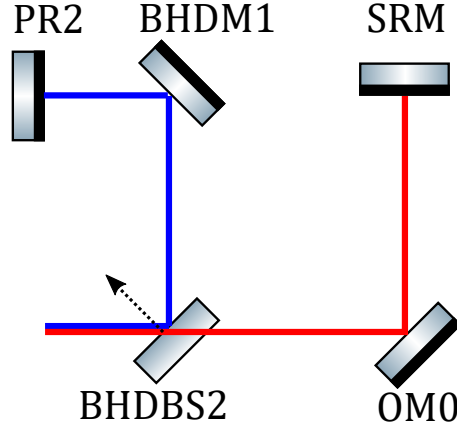


Figure 1.5: A sketch of the LO and signal beam paths. The LO beam path is shown in blue and the signal beam is shown in red. A dashed arrow indicates the direction that the recombining beam splitter, BHDBS2, may move in. If BHDBS2 moves in this way, the phase of the signal beam has increased while the phase of the LO beam has decreased at the point where the signal and LO beams recombine. This means that the recombination beam splitter motion needs to be counted twice in calculations of the phase noise requirement.

For a 98 mW LO beam and a 5 mW AS beam, the noise floor of the interferometer in units of LO path length is shown in figure 1.6. The requirement for the longitudinal motion of optics in the LO or signal path at 100 Hz is $5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$.

To determine the amount of isolation required, the typical motion of the optical table needs to be compared to the longitudinal motion requirement for the optics. A typical motion of the optical benches at 10 Hz is $10 \times 10^{-1} \text{m}/\sqrt{\text{Hz}}$ [13]. To achieve this with passive isolation, a triple suspensions will be used. These suspensions are known as ‘HAM Relay Triple Suspensions’ (HRTS).

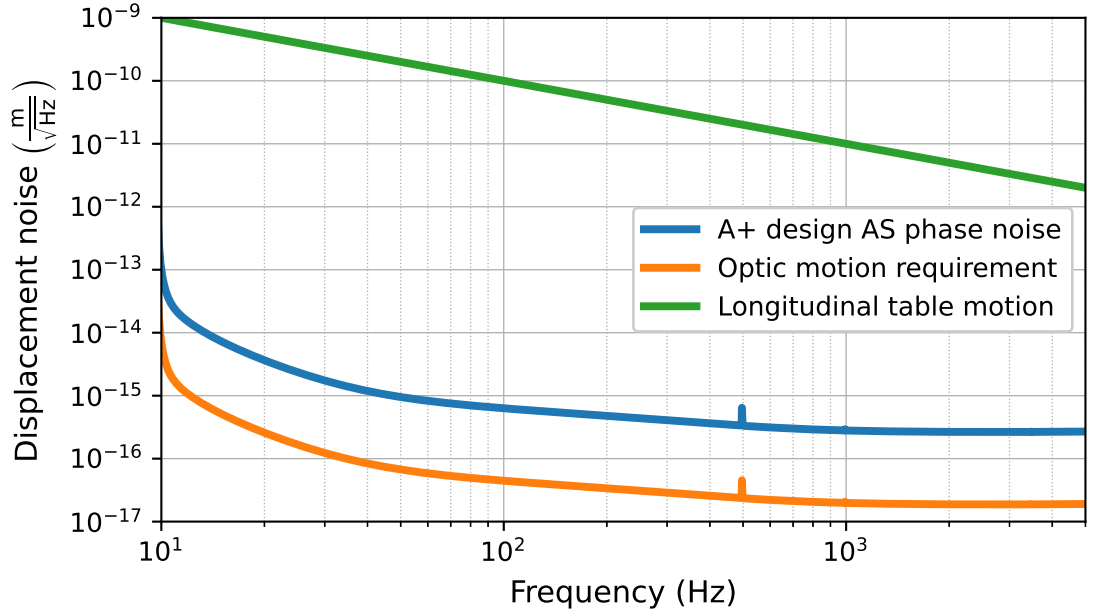


Figure 1.6: The phase noise between the LO and the AS beams causes noise. The sensitivity curve for A+ sets the limit on how much path length noise is acceptable between the AS and LO beams. The blue line shows sensitivity curve is shown in units of path length noise between the AS and LO beams, and the orange line shows the requirement of HRTS mirror motion. The table motion was modelled as $1/f$ noise.

1.3.4 Summary of LO path length Stability

Details of the LO path were discussed and analysed. The LO will be derived from the a beam picked off from the POP port, and a sketch of how the signal and LO will be combined is shown in figure 1.2. Starting with the linear response function between a differential strain and signal at the AS port, and the strain sensitivity curve for A+, we were able to derive a relative path length stability requirement for the local oscillator and signal beams, at 100 Hz this requirement is $5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$. By comparing this motion to a typical motion of the optical table, we were able to derive an isolation requirement, shown in Figure 1.6. This requirement led to triple suspensions being chosen to steer the LO and signal beams onto the combining beam splitter.

1.4 Conclusion

One of the major changes to LIGO for A+ will be the implementation of BHD. Combined with frequency dependent squeezing, BHD offers the possibility to optimise the detector for different gravitational wave sources. Along with this, BHD offers a way by which the coupling between the signal recycling cavity length noise and DARM can be reduced.

BHD presents some technical challenges over DC readout. The LO phase noise must be kept low enough as to not contaminate the signal at the AS port with LO path noise. The requirement for the phase noise is shown in figure 1.6. This requirement can be met by using triple suspensions to deliver to local oscillator and AS beams to the beam splitter they are combined on.

Chapter 2

Design of the HAM6 layout

The implementation of BHD requires a complete redesign of the HAM6 layout. Section 2 discusses the functionality needed from the HAM6 layout and analyses the proposed HAM6 layout. Section 2.3 examines the active wavefront control (AWC) that will be used to correct for mode mismatches from between the interferometer and the OMC. Section 2.4 concludes this chapter.

Fundamentally, the layout in HAM6 must combine the AS beam and the LO beam, and sense the combined beams to form a balanced homodyne detector. There are several other functionalities required and constraints imposed on the layout of HAM6. The following section will describe these functionalities and constraints and how they guided the design of the HAM6 layout. This work was undertaken by a team including myself, Ken Strain, Russel Jones, Stephen Webster and Mark Barton.

2.1 The Constraints and Requirements of the Layout of HAM6

This section will describe the functionality required from the the HAM6 layout, and some practical constraints that are imposed by the hardware and the infrastructure. The layout in HAM6 must provide steering and mode matching between the interferometer and the OMC. The main design challenge for this layout was to achieve 45° of Gouy phase between the mode matching optics, as this requires a lot of space. There are a few properties of the HAM6 layout that cannot be changed, such as the entry point of beams and size of the optical table.

As the constraints give a starting point for all designs, they will be discussed first¹. These include: where beams enter the HAM6 chamber, the size of the OMC waist, and the size of nominal mode entering HAM6. The size of the HAM optical table does restrict the design, however small overhangs sometimes are permitted as table extensions can be made.

The entry point of the AS and LO beams in principle can be anywhere in HAM6, however in reality they are fixed due to the location of HAM6 with respect to the interferometer, due to the location of the septum windows between HAM5 and HAM6, and due to the many other components of the interferometer that these beams need to avoid. The LO beam is steered with BHDM1 onto BHDBS2, and

¹It could be argued that requirements are a constraint in a design. Here, the constraints are geometric properties which we have no freedom in, such as the input beam shape, the OMC geometry and the layout of the rest of the rest of the interferometer. A requirement, such as the desired minimum mode matching, is a property which doesn't have a set 'physical form'. For instance, while Gouy phase does correspond to a physical distance, not all 45° Gouy phase spaces are the same physical size.

the AS beam is steered with OM0 onto BHDBS2. This is required to provide the degrees of freedom to co-align the AS and LO beams. There is some freedom in choice for the location of OM0 and BHDBS2, but there is no choice in the location of the AS and LO beams.

The design beam parameter at the SRM is $q_{\text{SRM}} = -3.55 + 1.09i$ m, so the design of the HAM6 mode matching optics is based on this number. In reality, the beam parameter could be different to this, as discussed in chapter ??). At the SRM, there are two sets of modes we could design the AWC to match to: The arm mode, or the SRC mode. Ideally, these would be the same mode, however geometry errors in the SRC cause the arm and the SRC to be mismatched meaning the radius of curvature and beam size at the SRM will not correspond to the eigenmode of the SRM. This set of likely modes, shown in ??, was taken into consideration when choosing a layout for the AWC in HAM6.

The current OMC design (with trivial modifications) will be used to remove junk light from the signal beam, so there is no choice in target waist size. The input beam at the SRM is defined in T1200410. The mode matching is designed assuming that the beam at the SRM is the ‘design mode’. Mode matching between the beam exiting the SRM and the OMC waist will be done with a static lens and 2 active mirrors. The static lens will be placed just after the OFI and its purpose is to provide sufficient Gouy phase separation of the two active mirrors.

It was decided that the two beams after BHDBS2 would have the same path length to reduce the cost of the HAM6 optics. This may also improve the common-mode noise rejection ratio.

The requirements for the HAM6 layout are to provide steering for the interferometer beam into the OMC, and mode matching between the interferometer beam and the OMC. The goal for mode matching is 98%. After BHDBS2 the layout must have two mirrors separated 90° and 45° of Gouy phase to provide orthogonal control over the beam pointing and mode-matching. The 45° Gouy phase separation requirement is what makes the design of HAM6 none trivial.

The goal for mode matching between the arm mode and the OMC mode is is $> 98\%$. The SAMS actuators will provide ± 100 mD of optical power tuning range, with a bandwidth of the order 100 Hz. Astigmatism can be considered as a loss, and none-zero angles of incidence (AOI) of beams onto curved optics results in a the beam becoming astigmatic. This prevents designs using large AOI. This is described in detail in Chapter ??.

Nominally, the role of the AWC in HAM6 is to correct for geometry errors of the optics after the SRM such that the mode matching between the Arm mode and the OMC waist is maximised. This assumes that the changes in the interferometers mode, for instance due to a change in thermal state, will be corrected for by other subsystems e.g. the TCS. In practice, operators of the interferometer do what ever improves the SNR of the the gravitational wave signal. For instance, it has been found that TCS can be used to decrease laser intensity noise coupling to the the GW signal channel. The AWC in HAM6 was designed with this flexibility in mind. The losses due to components in HAM6 is summarised in table 2.1 (See E1900385).

While in principle only two curved mirrors would be required to mode match the beam from the interferometer to the OMC, in fact a single mirror may be able to do

that, a lens in the OFI will be used to provide further Gouy phase separation between the two active mode matching mirrors after BHDBS2. An active lens was considered here, but the unavailability of flint glass meant that this idea was discarded. There were also concerns about the heating of the OFI assembly that an active lens may cause. As well as this, the range of optical powers for the SAMS is large enough that it would be an over-design to implement an active lens in the OFI.

To provide orthogonal mode matching control, two of the mirrors should be separated by 45° . This Gouy phase separation requirement give us a method of generating candidate designs, as these can be interpreted as needing the mode matching mirrors to be as far apart as possible from each other on the HAM table. As the Rayleigh range of the beams is on the order of a meter, and most gouy phase is accumulated within one Rayleigh range either side of a beam waist, a beam will have to travel quite far across the table before it acquires sufficient Gouy phase.

Additionally, To provide orthogonal alignment control in each beam path after BHDBS2, two of the mirrors are required to be separated by 90° of Gouy phase. This was always done with the mirror immediately after BHDBS2 and the mirror immediately before the OMC. It was found that placing the first mirror close to BHDBS2 often provided the best layouts.

Light scatter from the OMC is considered a potential source of noise. The OMCs are expected to be the largest contributor of back-scatter from HAM6 back to the interferometer. By having the OMCs after the AS and LO beams are combined, and by having the OFI protect the interferometer, the requirement on the motion of the OMCs is fairly lax. Nevertheless, a single stage suspended platform is required.

This will provide isolation down to 4 Hz.

Additional sensors in HAM6 are required for controlling various degrees of freedom within the layout. Some of these degrees of freedom will be monitored using beams which are picked off from the main beam path, and so additional mirrors will be required to steer these beams around the chamber. A set of smaller optics will be installed to provide diagnostic beams which can be imaged with CCDs. A shutter will be required to protect the hardware in HAM6 from being damaged by the large amounts of light that can potentially be transmitted through the SRM. Appendix ?? describes how the longitudinal and pointing degrees of freedom may be sensed and controlled.

In summary, the main constraints for the design of HAM6 are where the beams enter the chamber, the size of the optical bench, the input mode and the OMC design. There must be low amounts of loss, meaning large AOI should be avoided. The reason the layout of HAM6 is a none trivial design task is it is desirable to have orthogonal control over the modematching and beam direction. This requires a long optical axis of a similar to the size of a HAM table. In addition to this, the HAM6 layout needs to leave space for all the other sensors and beams used for control. This means that all design will need nearly all of the table space available.

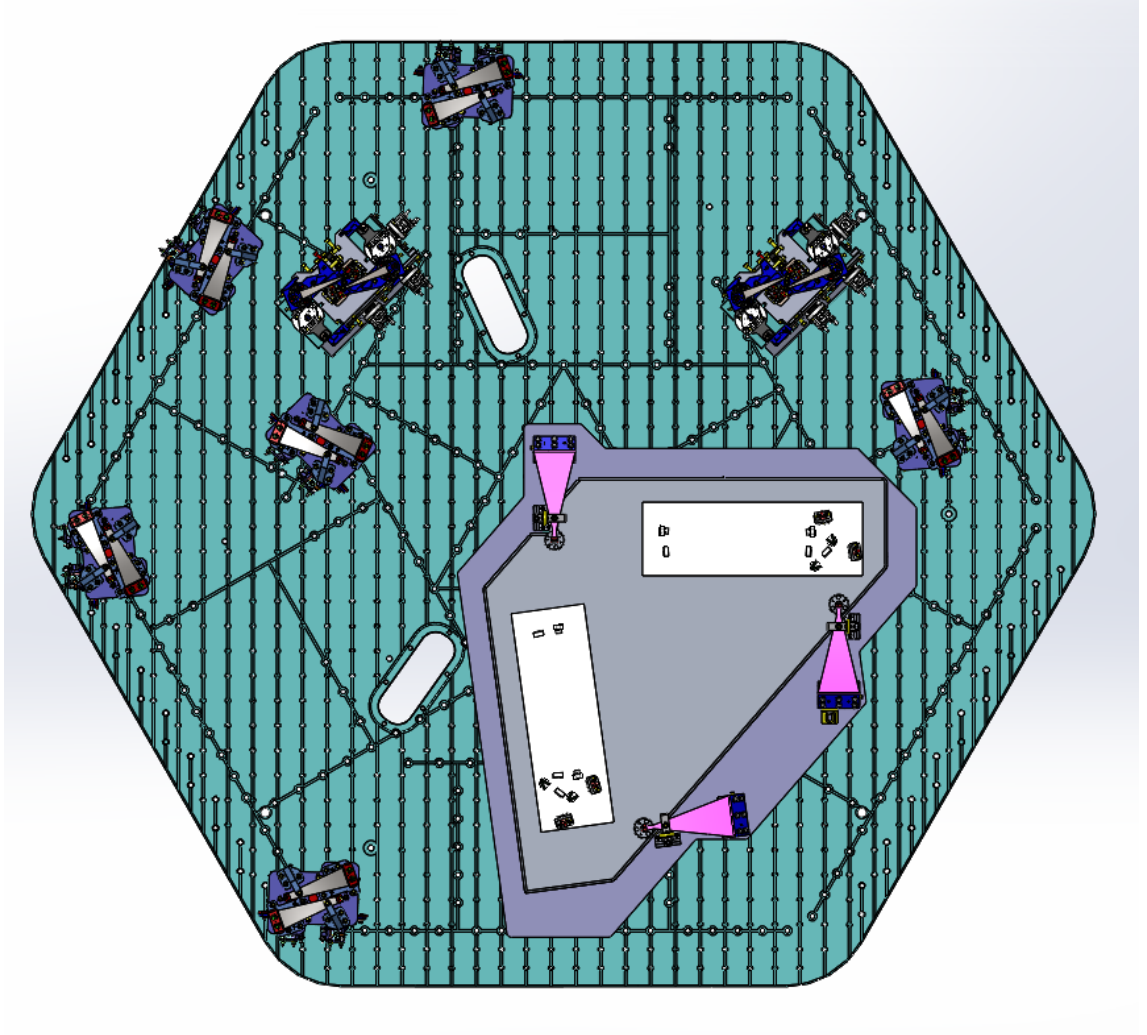


Figure 2.1: Place holder graphic. The pentagram layout. CAD with beam path will be inserted later.

HAM6 loss Source	Target for O5
OMC throughput	$< 2\%$
OM3 transmission (or equivalent) for sensing	1–2%
PD quantum efficiency	1%
Mode mismatch to the OMC	$< 2\%$ (as low as possible)
Any other optical loss	$< 1\%$

Table 2.1: Target for losses in HAM6. Reproduction of E1900385

2.2 Chosen Layout

The design that is being taken forward at the time of writing is known as the ‘Pentagram’², and is shown in figure 2.1. Optically, there are many layouts which could have satisfied the above requirements and constraints. Ultimately the choice of layout for HAM6 was determined by mechanical factors such as table space and number of suspensions. The optical properties of this layout are summarised in table 2.2.

The beam size and Gouy phase as the a function of position along the optical axis is shown in figure 2.2. The positions of significant points along the optical axis is also shown. The size of the beam at the final mode matching optic matches the size of the target mode, as the size of the beam cannot be changed at a mirrors surface. The beam is always less than 1.2 mm within HAM6.

The astigmatism loss was calculated to be $> 0.05\%$. This astigmatism loss was plotted as a function of angle, and above 10° the loss becomes significant. The compromise that is made when choosing a larger AOI is the ability to have more Gouy phase separation between OMx1 and OMx2. In reality, the IFO and OMC will also

²It was called this as when viewed from above the main beam path looks like a pentagram. An alternative and much less interesting name for this layout is ‘5f’.

have a small amount astigmatism. Losses of $> 0.05\%$ are considered unimportant, especially when compared to the other expected losses, some of which are expected to be at the many percent level, in the readout of the interferometer.

The focal lengths of the OFI lens, OMx1 and OMx2 were optimised so that there was a close to 45° Gouy phase separation between OMx1 and OMx2 as possible. This is shown in figure 2.3. With this layout, it was possible to achieve 43° Gouy phase separation between OMx1 and OMx2. Other layouts were able to provide exactly 45° Gouy phase separation, however the Pentagram layout offers many practical benefits over the other considered designs that outweigh an additional 2° of Gouy phase separation. This layout also provides good Gouy phase separation for alignment control, with there being 88° between OMxS and OMx2.

In conclusion, a HAM6 layout that fulfils the criteria set in Section 2.1 has been designed. This layout is shown in Figure 2.1. Near optimal mode-matching control and alignment control was achieved, and all the other functionality required from the layout have been considered. The next stage of development for this layout involves the design and prototyping of parts, and the consideration of fine details such as cable routing.

Optic	Power (1/m)	Focal length (m)	RoC (m)
OFI lens	-0.0583	-17.1552	-
OMx1	-0.3539	-2.8290	5.6580 (concave)
OMx2	-0.8737	-1.1445	2.2891 (concave)

Space Name	Optical Path Length (mm)
SRM to OFI lens	1120
OFI lens to HAM6 table edge	720
HAM6 table edge to OM0	432
OM0 to BHDBS2	905
BHDBS2 to OMxS	225
OMxS to OMx1	1240
OMx1 to OMx2	1590
OMx2 to OMCx waist	1220

Parameter	Value
SRM Complex Beam Parameter	$q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$
OMC waist	$485 \text{ }\mu\text{m}$
OMx1 to OMx2 Gouy phase	43°
OMxS to OMx2 Gouy phase	88°
OMxS and OMx2 AOI	10°

Table 2.2: Mechanical and optical properties of the Pentagon layout.

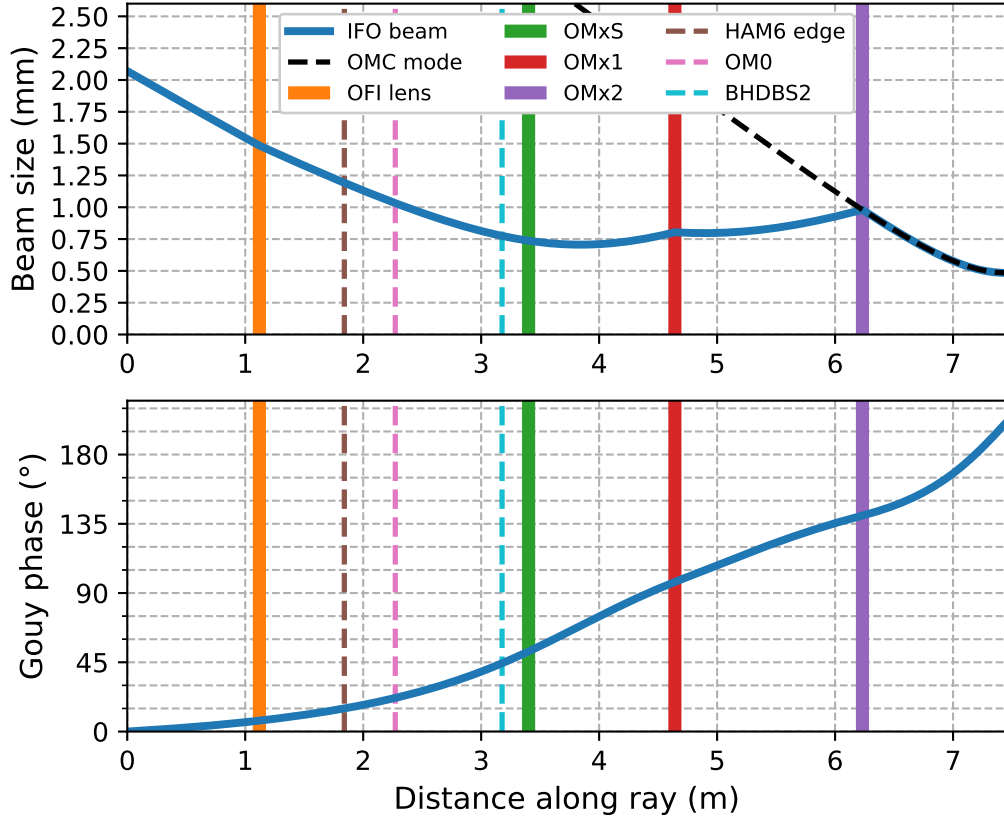


Figure 2.2: Top panel: the radius of the beam as it travels from the SRM ($z = 0$) to the waist of the OMC. Significant points along the axis are marked with vertical lines. Less significant, but useful reference points, are marked by dashed vertical lines. Bottom panel: The gouy phase shift accumulated by the beam as it propagates along the optical axis.

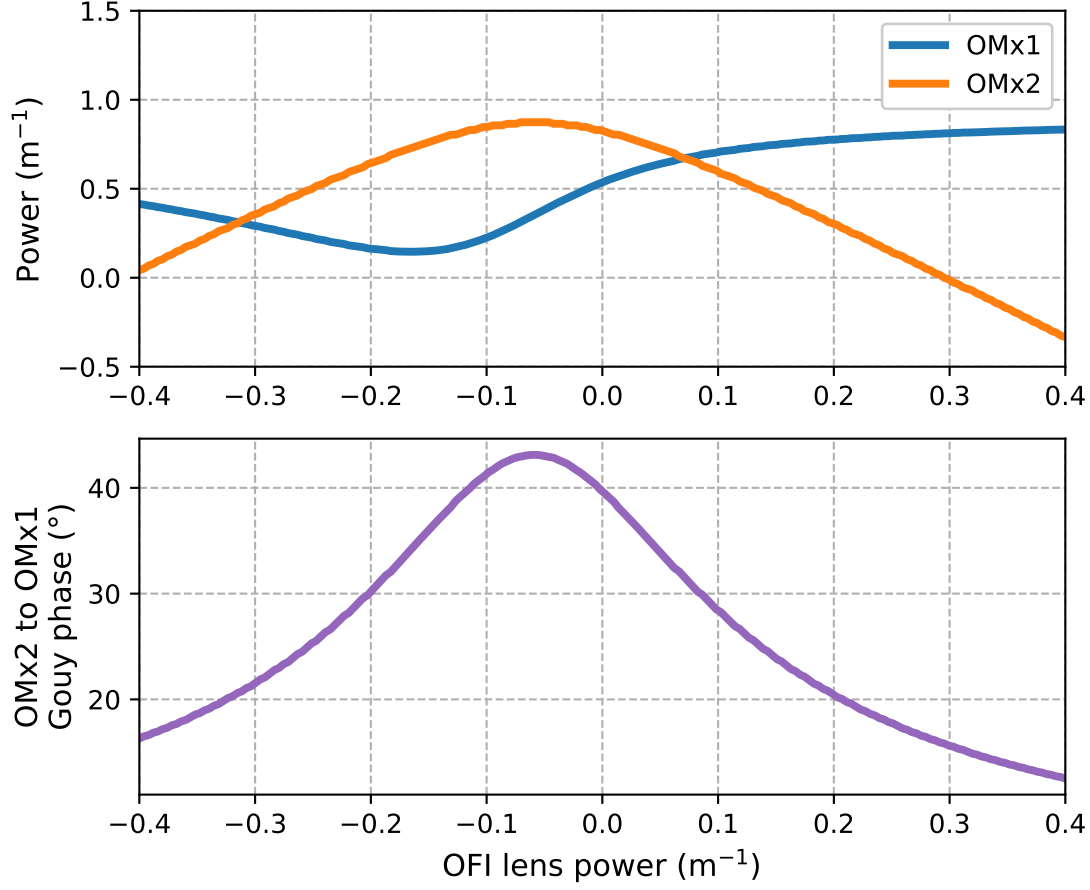


Figure 2.3: Top panel: The choice of OFI lens power effects the combination of OMx1 and OMx2 power required for 100% mode matching. Bottom panel: The best choice of OFI lens corresponds to where the Gouy phase separation between the pair of OMx1 and OMx2 mirrors that give 100% mode matching is closest to 45° . The optimal Gouy phase separation between OMx1 and OMx2 achievable by this layout is 43° .

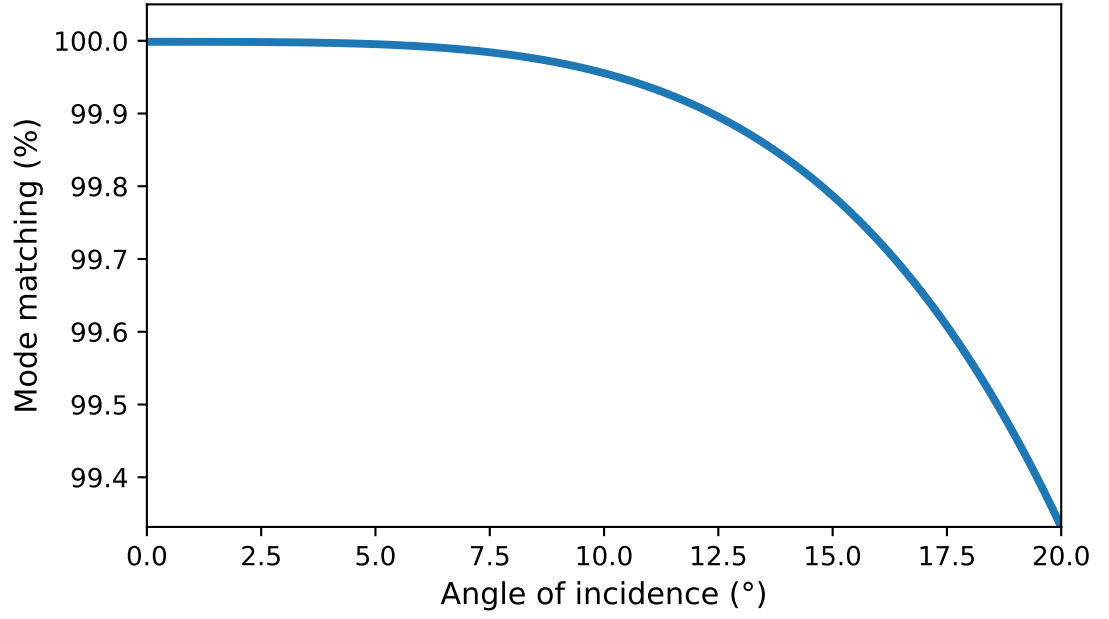


Figure 2.4: Astigmatism is a source of loss. When designing this layout, to explore how sensitive it was to astigmatism, the AOI for the curved mirrors was altered. Using equation 2.1, the loss was computed. This effect is negligible up to 10° , after which it starts to become significant compared to the losses summarised in table 2.1.

2.3 WS visualisations for Analysing the Robustness of the HAM6 Layout to Mode-Mismatch

It is expected that there will be some mismatch between the beam from the interferometer and the OMC due to geometry errors within the interferometer. To visualise the effectiveness of the mode matching control provided by the active optics, it is useful to consider a 2D space representing Gaussian beams. The region of this space that can be mode matched by the proposed layout and a distribution of modes that the layout may have to correct for can be visually compared to gain insight into the effectiveness of the layout. It was found that the proposed layout described in 2.1 will be cover almost all expected input beams.

The mode-matching between two laser beam modes, or a laser mode and a cavity mode, describes how similar those two modes are. In general, the two modes can be mismatched by being misaligned or by having a different beam shape. Mode matching typically refers to the overlap of the two modes if they were perfectly aligned.

Mode-matching is usually, to first approximation, thought of as a difference in beam waist size and beam size location. This can be mathematically described by the $\text{HG}_{20} + \text{HG}_{02}$ mode. The effect of changing the waist location of the mode is equivalent to an in-phase change in HG_{20} content, whereas an change in waist location is orthogonal to this. [read up on this](#). Sensing the $\text{HG}_{20} + \text{HG}_{02}$ mode therefore gives a measure of the mode-matching between two lasers. As the $\text{HG}_{20} + \text{HG}_{02}$ acquires 3ϕ compared to a plane wave, and 2ϕ compared to the HG_{00} mode,

spacing the actuators at 45° will give orthogonal control of the mode matching. This means that the beam size and beam waist can be independently controlled.

Two perfectly mode matched beams will have identical beam parameters. If the input mode and the target mode differ in beam size and defocus, then some light will be scattered into the $\text{HG}_{20} + \text{rmHG}_{02}$ mode. The amount of mode matching $\mathcal{M}$ between two astigmatic modes, $q_{1,2}$, is

$$\mathcal{M} = \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (2.1)$$

For a stigmatic beam, set $q_x = q_y$. The mode matching can be calculated at any point along the optical axis of the layout, as the mode matching must be preserved, but it makes sense to choose a physically significant plane such as the input plane, a plane containing a curved optic, or the plane containing the waist of the target beam.

A curved mirror can only change the curvature of a Gaussian beam, meaning that at the mirrors surface the beams before and after reflection must have the same size. The process of a Gaussian beam described by q_1 being reflected from a mirror with power S_m to produce a new Gaussian beam, q_2 , can be written as

$$\frac{1}{q_2} = S_m + \frac{1}{q_1} = (S_m + S_1) - i \frac{\lambda}{\pi w_1^2}. \quad (2.2)$$

To visualise how a mode is changed by an optical layout, beam parameters can be represented in a 'WS space'. This space parametrises a beam in terms of its size, w , and its defocus, S . This has the advantage of showing the beam in terms of

practical quantities. The downside to this representation is as the beam propagates through the layout, the path that is traced through the WS space by the mode is fairly complex.

There are two other ways that a beam parameter could be graphically represented. As q is a complex number, the beam parameter could be represented in terms of its real and imaginary components. This makes seeing the effect of propagation straight forward, but the effect of a focussing element is harder to visualise. Alternatively, plotting $1/q$ in terms of $\lambda/\pi w^2$ and s produces plots where the effect of a focussing elements and propagation is easy to visualise, as focussing is just a change in defocus and propagation follows a circular path. However, this technique obscures the beam size, and as these graphs are being used to visualise the beam, rather than design the optical layout³, the WS representation was chosen.

To demonstrate how orthogonal actuators can be represented in WS space, consider the two lenses with variable power separated by 45° of Gouy phase. As the two lenses are orthogonal, the waist size and waist location of the mode exiting this layout will orthogonally controllable. In a WS space, the orthogonality of two actuators can be seen by the angle between the two vectors that represent the actuation. If the two lenses are orthogonal, then the actuation vectors will be orthogonal. Figure 2.5 is a sketch of the WS space representation for a simple optical layout consisting of two active lenses separated by 45° Gouy phase. If there is large amounts of astigmatism, it no longer makes sense to represent the beam with just one WS space diagram.

³This representation is useful when designing an optical layout by hand, something which is not often done anymore.

The mode matching requirement for LIGO is $> 98\%$. It is useful to plot contours of the mode overlap in the WS space between a mode and a target mode. Note that these contours can only tell you about the overlap between a mode and the target mode, the mode matching between two arbitrary modes can not be deduced from these contours. An example of a situation where this is obvious is two different modes may be, say, 90% mode matched to the target. As they are different from each other, they cannot be 100% mode matched.

To visualise the effectiveness of the actuation provided by the active elements of the HAM6 layout, the region of the WS from which the target mode matching space achieved was computed. To do this, first the ring of modes in WS space that is 98% matched after reflection from the second active mirror, OMx2, was considered. These modes must be within $S_{\text{OMx2}} \pm \Delta S$, where S_{OMx2} is the static power of OMx2 and ΔS is the actuation range of the SAMS. If OMx2 had no actuation, then the effect of OMx2 would be to move the ring vertically in the WS plane. However, as power of OMx2 can be increased or decreased, the shape of the ring of modes that can be mode matched to 98% can be changed.

Crudely, the shape of this space can be seen by applying $S_{\text{OMx2}} \pm \Delta S$ to the 98% ring, producing 2 new rings. If the ring is cut by a line which intersects the locations on the ring at the extrema of the beam size, then two regions of the 98% ring can be created. For one of these regions, to maximise the amount of WS space covered by the actuator, the actuation applied should be $+\Delta S$, whereas for the other region, the actuation should be $-\Delta S$. This gives a set of modes which can be propagated to the first mirror, OMx1, where the same procedure can be applied. This process

shown in Figure 2.6.

The modes from the SRM that can be matched at to the target is shown in Figure 2.7. The visualise how successful the layout is at the handling likely modes from the arm described in Chapter ??, these modes were compared to the region of WS phase space from which the 98% mode matching requirement from a starting point of 92% mismatch. It was found that almost all likely modes will be covered by the layout, as the modes near the edge of this continuum are more unlikely to occur than the ones near the centre.

For all candidate layouts analysed, there were no major difference in the amount of modes that could be mode matched. Its worth noting that the in figure 2.7, modes at the yellow end of the spectrum may not be physical as they correspond to an unstable SRC. In principle, the requirement for having 45° of Gouy Phase separation between OMx1 and OMx2 doesn't maximise the amount of interferometer modes that can be mode matched. The shape that the purple region will have when HAM6 is constructed will depend on the optical powers of OMx1 and OMx2,

Nevertheless, Figure 2.7 is encouraging since it shows that the actuation provided by the OMx1 and OMx2 is very forgiving, so much so that the strict definition of the role of AWC in HAM6 may be relaxed in order for operators to use the AWC here to correct for other errors in the geometry of LIGO.

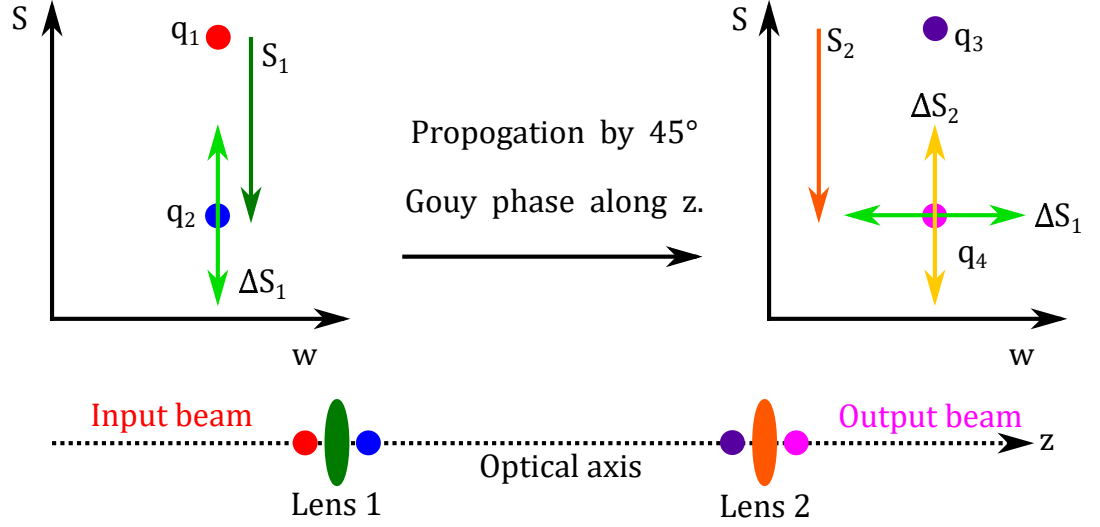


Figure 2.5: Sketches of the WS representation for a Gaussian beam interacting with different parts of a simple optical layout consisting of two active lenses with powers $S_{1,2} + \Delta S_{1,2}$. The left hand axis shows the WS plane at lens 1. The input beam q_1 is indicated by a red dot, and the effect of the lens on q_1 is shown by the dark green arrow S_1 . This results in a new mode q_2 , which is shown as a blue dot. A light green arrow indicates the actuation of lens 1. The right hand axis shows the WS plane at lens 2. The input mode at lens 2, q_3 , is shown as a purple dot, and the output mode, q_4 , is shown as a pink dot. The static and dynamic power of lens 2 is indicated by the orange and yellow lines. As lens 1 and lens 2 are separated by 45° of Gouy Phase, the actuation provided by lens 1 results in a change in the beam parameter which is orthogonal to the actuation provided by lens 2, provided that the actuations are small enough.

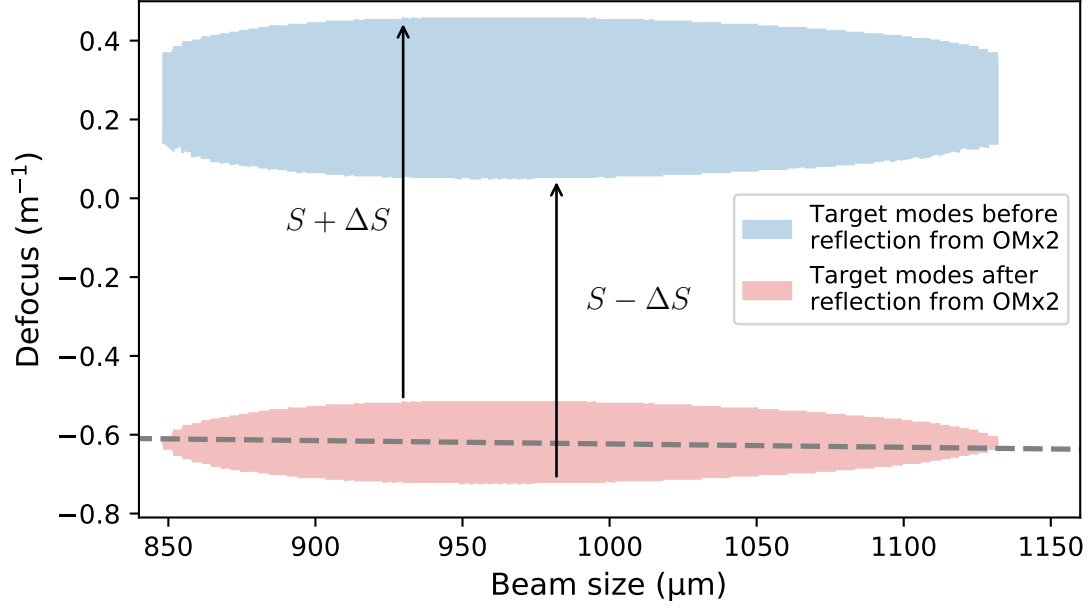


Figure 2.6: This is the WS space at the OMx2 plane. The effect of OMx2 in the plane of OMx2 is to move a mode vertically in WS space. After reflection from OMx2, a mode must be at least 98% mode matched to the OMC as there are no further active mode matching elements in the layout of HAM6. Modes with at least 98% are shown by the red region. The blue region represents which modes can be reached from the red region by either $S_{\text{OMx2}} \pm \Delta S$, and was constructed by applying transformations to the modes in the red region. The red region was split into two parts, ‘upper’ and ‘lower’, indicated by the dashed line that connects the two extremes in width of the region. To create the lower edge of the blue region, modes in the lower edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} - \Delta S$ applied to them. To construct the upper edge of the blue region, upper edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} + \Delta S$ applied to them.

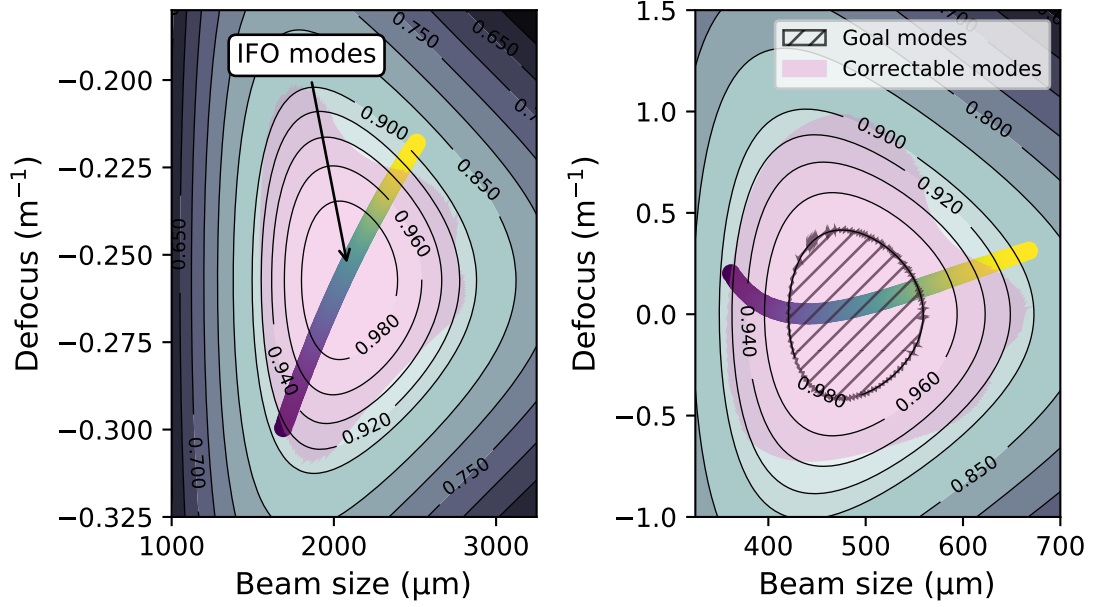


Figure 2.7: Left panel: the set of modes that could be present at the SRM is shown by the purple-blue-green-yellow curve, as described in chapter ???. The pink region shows which modes can reach 98% mode matching by using the the AWC in HAM6. The contours indicate the mode matching between a mode (W, S) and the OMC mode. Right panel: this shows the same modes as the top panel, except they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region, which is indicated with a hatched area.

2.4 Conclusion about BHD for A+

The functionality required from the BHD layout was discussed, and a proposed layout that satisfies these requirements is shown in figure 2.1. The optical properties of this layout were investigated, and by selection of the right combination of mode matching optics, near orthogonal control of the AS-LO beam alignment and mode matching to the OMC was achieved.

By plotting the $< 98\%$ mode matching requirement in WS space, the robustness to changes in the input beam parameter was investigated. The AWC provides a huge amount of mode matching range that the layout can tolerate very large changes in parameter at the SRM. It was found that almost all possible beams will be able to mode matched to the OMC. This is shown in figure 2.7.

Appendix A

Quantum Optics

[1]

A.1 Definitions

Beam splitter transforms as

$$c = \frac{1}{\sqrt{2}}a + \frac{1}{\sqrt{2}}b \quad (\text{A.1})$$

$$d = \frac{1}{\sqrt{2}}a - \frac{1}{\sqrt{2}}b \quad (\text{A.2})$$

$$a = \frac{1}{\sqrt{2}}c + \frac{1}{\sqrt{2}}d \quad (\text{A.3})$$

$$b = \frac{1}{\sqrt{2}}c - \frac{1}{\sqrt{2}}d \quad (\text{A.4})$$

$$(\text{A.5})$$

Input field is $|\Psi_{\text{in}}\rangle$, output field is $|\Psi_{\text{out}}\rangle$. Vacuum = $|\vec{0}\rangle = |0\rangle_a |0\rangle_b |0\rangle_c |0\rangle_d$.

Coherent state:

$$|\alpha\rangle_a = \exp(\alpha a^\dagger - \alpha^* a) |0\rangle_a. \quad (\text{A.6})$$

Coherent states are not orthogonal, $\langle\alpha|\beta\rangle \neq \delta(\alpha - \beta)$

Displacement operator $D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a)$

Squeezed state:

$$|\xi\rangle_a = \exp\left(\frac{1}{2}\xi^* a^2 - \frac{1}{2}\xi a^{\dagger 2}\right) |0\rangle_a. \quad (\text{A.7})$$

Squeeze operator $S(\xi) = \exp\left(\frac{1}{2}\xi^* a^2 - \frac{1}{2}\xi a^{\dagger 2}\right)$.

Expectation $\langle\hat{O}\rangle = \langle\Psi|\hat{O}|\Psi\rangle$ and $\Delta O = \sqrt{\langle\hat{O}^2\rangle - \langle\hat{O}\rangle^2}$.

Difference operator: $j = c^\dagger c - d^\dagger d$

A.2 single photon state and a beam splitter

Input field:

$$|\Psi_{\text{in}}\rangle = |\Psi_{\text{in}}\rangle = |1\rangle_a |0\rangle_b |0\rangle_c |0\rangle_d \quad (\text{A.8})$$

$$= a^\dagger |\vec{0}\rangle. \quad (\text{A.9})$$

Output field:

$$|\Psi_{\text{out}}\rangle = \left(\frac{1}{\sqrt{2}}c + \frac{1}{\sqrt{2}}d\right)^\dagger |\vec{0}\rangle. \quad (\text{A.10})$$

$$= |0\rangle_a |0\rangle_b \left(\frac{1}{\sqrt{2}}|1\rangle_c |0\rangle_d + \frac{1}{\sqrt{2}}|0\rangle_c |1\rangle_d\right). \quad (\text{A.11})$$

This is an entangled state. At the beam splitter, the photon has an equal probability of being transmitted or reflected.

A.3 2 Single photon states entering the beam splitter

Input field:

$$|\Psi_{\text{in}}\rangle = |\Psi_{\text{in}}\rangle = |1\rangle_a |1\rangle_b |0\rangle_c |0\rangle_d \quad (\text{A.12})$$

$$= ab |\vec{0}\rangle. \quad (\text{A.13})$$

Output field:

$$|\Psi_{\text{out}}\rangle = \left(\frac{1}{\sqrt{2}}c + \frac{1}{\sqrt{2}}d \right)^\dagger \left(\frac{1}{\sqrt{2}}c - \frac{1}{\sqrt{2}}d \right)^\dagger |\vec{0}\rangle \quad (\text{A.14})$$

$$= \left(\frac{1}{2}c^{\dagger 2} - \frac{1}{2}d^{\dagger 2} \right) |\vec{0}\rangle \quad (\text{A.15})$$

$$= |0\rangle_a |0\rangle_b \left(\frac{1}{\sqrt{2}} |2\rangle_c |0\rangle_d + \frac{1}{\sqrt{2}} |0\rangle_c |2\rangle_d \right). \quad (\text{A.16})$$

This is known as the Hong-Ou-Mandel effect. This is interference between the two ways of making a $|0\rangle_a |0\rangle_b |1\rangle_c |1\rangle_d$ state.

A.4 coherent state and a beam splitter

Input field:

$$|\Psi_{\text{in}}\rangle = |\alpha\rangle_a |0\rangle_b |0\rangle_c |0\rangle_d \quad (\text{A.17})$$

$$= \exp(\alpha a^\dagger - \alpha^* a) |\vec{0}\rangle. \quad (\text{A.18})$$

Output field:

$$|\Psi_{\text{out}}\rangle = \exp\left(\alpha\left(\frac{1}{\sqrt{2}}c + \frac{1}{\sqrt{2}}d\right)^\dagger - \alpha^*\left(\frac{1}{\sqrt{2}}c + \frac{1}{\sqrt{2}}d\right)\right) |\vec{0}\rangle \quad (\text{A.19})$$

$$= |0\rangle_a |0\rangle_b \left|\frac{\alpha}{\sqrt{2}}\right\rangle_c \left|\frac{\alpha}{\sqrt{2}}\right\rangle_d. \quad (\text{A.20})$$

This is similar to a classical field. If the initial state was $|\Psi_{\text{in}}\rangle = |0\rangle_a |\beta\rangle_b |0\rangle_c |0\rangle_d$, then the output would be $|\Psi_{\text{out}}\rangle = |0\rangle_a |0\rangle_b \left|\frac{\beta}{\sqrt{2}}\right\rangle_c \left|-\frac{\beta}{\sqrt{2}}\right\rangle_d$.

A.5 2 coherent states and a beam splitter

Input field:

$$|\Psi_{\text{in}}\rangle = |\alpha\rangle_a |\beta\rangle_b |0\rangle_c |0\rangle_d \quad (\text{A.21})$$

$$= \exp(\alpha a^\dagger - \alpha^* a) \exp(\beta b^\dagger - \beta^* b) |\vec{0}\rangle. \quad (\text{A.22})$$

Output field:

$$|\Psi_{\text{out}}\rangle = \exp\left(\alpha\left(\frac{1}{\sqrt{2}}c + \frac{1}{\sqrt{2}}d\right)^\dagger - \alpha^*\left(\frac{1}{\sqrt{2}}c + \frac{1}{\sqrt{2}}d\right)\right) \quad (\text{A.23})$$

$$\times \exp\left(\beta\left(\frac{1}{\sqrt{2}}c - \frac{1}{\sqrt{2}}d\right)^\dagger - \beta^*\left(\frac{1}{\sqrt{2}}c - \frac{1}{\sqrt{2}}d\right)\right) |\vec{0}\rangle \quad (\text{A.24})$$

$$= \exp\left(\frac{\alpha + \beta}{2}c^\dagger - \frac{\alpha^* + \beta^*}{2}c\right) \exp\left(\frac{\alpha - \beta}{2}d^\dagger - \frac{\alpha^* - \beta^*}{2}d\right) |\vec{0}\rangle \quad (\text{A.25})$$

$$= |0\rangle_a |0\rangle_b \left| \frac{\alpha + \beta}{\sqrt{2}} \right\rangle_c \left| \frac{\alpha - \beta}{\sqrt{2}} \right\rangle_d. \quad (\text{A.26})$$

Selecting the phase between a and b controls how much light is at c and d . Setting $\alpha = \beta$ results in destructive interference at d and constructive at c . Setting $i\alpha = \beta$ maximises the spreads the light evenly in both c and d .

The expectation for i is found by measuring the c port and d port of the beam splitting with photo diodes and subtracting the results:

$$\langle \Psi_{\text{out}} | c^\dagger c - d^\dagger d | \Psi_{\text{out}} \rangle = \langle \Psi_{\text{out}} | \left(\frac{\alpha + \beta}{\sqrt{2}} \right)^* \left(\frac{\alpha + \beta}{\sqrt{2}} \right) | \Psi_{\text{out}} \rangle \quad (\text{A.27})$$

$$- \langle \Psi_{\text{out}} | \left(\frac{\alpha - \beta}{\sqrt{2}} \right)^* \left(\frac{\alpha - \beta}{\sqrt{2}} \right) | \Psi_{\text{out}} \rangle \quad (\text{A.28})$$

$$\langle j \rangle = \alpha^* \beta + \alpha \beta^*. \quad (\text{A.29})$$

The phases can be set such that α is real and β can be written as $|\beta| \exp(i\theta)$. Equation (A.29) may be written as

$$\langle j \rangle = \alpha |\beta| \exp(i\theta) + \alpha |\beta| \exp(-i\theta) \quad (\text{A.30})$$

$$= 2\alpha |\beta| \cos(\theta). \quad (\text{A.31})$$

Comparing equation (A.29) to equation (1.14), $P = 2|E_s E_{\text{LO}}^\dagger| \cos(\phi)$, we see that the quantum expectation for the BHD signal when two coherent states are used has the same form as its classical counterpart.

To find Δj , first we need to calculate $\langle i^2 \rangle$,

$$\begin{aligned}
\langle i^2 \rangle &= \langle \Psi_{\text{out}} | (c^\dagger c - d^\dagger d)^2 | \Psi_{\text{out}} \rangle \\
&= \langle \Psi_{\text{out}} | c^\dagger c c^\dagger c + d^\dagger d d^\dagger d | \Psi_{\text{out}} \rangle - 2 \langle \Psi_{\text{out}} | c^\dagger c d^\dagger d | \Psi_{\text{out}} \rangle \\
&= \langle \Psi_{\text{out}} | c^\dagger (c^\dagger c + 1) c + d^\dagger (d^\dagger d + 1) d | \Psi_{\text{out}} \rangle - 2 \langle \Psi_{\text{out}} | c^\dagger c d^\dagger d | \Psi_{\text{out}} \rangle \\
&= \langle \Psi_{\text{out}} | c^\dagger (c^\dagger c + 1) c + d^\dagger (d^\dagger d + 1) d | \Psi_{\text{out}} \rangle - 2 \langle \Psi_{\text{out}} | c^\dagger c d^\dagger d | \Psi_{\text{out}} \rangle \\
&= \left\{ \left(\frac{\alpha + \beta}{\sqrt{2}} \right)^* \left(\frac{\alpha + \beta}{\sqrt{2}} \right) \right\}^2 + \left(\frac{\alpha + \beta}{\sqrt{2}} \right)^* \left(\frac{\alpha + \beta}{\sqrt{2}} \right) \\
&\quad + \left\{ \left(\frac{\alpha - \beta}{\sqrt{2}} \right)^* \left(\frac{\alpha - \beta}{\sqrt{2}} \right) \right\}^2 + \left(\frac{\alpha - \beta}{\sqrt{2}} \right)^* \left(\frac{\alpha - \beta}{\sqrt{2}} \right) \\
&\quad - 2 \left(\frac{\alpha + \beta}{\sqrt{2}} \right)^* \left(\frac{\alpha + \beta}{\sqrt{2}} \right) \left(\frac{\alpha - \beta}{\sqrt{2}} \right)^* \left(\frac{\alpha - \beta}{\sqrt{2}} \right) \\
&= 0,
\end{aligned} \tag{A.32}$$

where the substitutions

$$\left(\frac{\alpha + \beta}{\sqrt{2}} \right)^* \left(\frac{\alpha + \beta}{\sqrt{2}} \right) = \frac{\alpha^2}{2} + \frac{|\beta|^2}{2} + \alpha |\beta| \cos(\theta) \tag{A.33}$$

$$\left(\frac{\alpha - \beta}{\sqrt{2}} \right)^* \left(\frac{\alpha - \beta}{\sqrt{2}} \right) = \frac{\alpha^2}{2} + \frac{|\beta|^2}{2} - \alpha |\beta| \cos(\theta) \tag{A.34}$$

are helpful.

Therefore, the variance of the signal is

$$\Delta j = i2\alpha |\beta| \cos \theta. \quad (\text{A.35})$$

The relationship between homodyne phase fluctuations and the noise in the signal is

$$\Delta \theta = \Delta j \left/ \left| \frac{\partial \langle j \rangle}{\partial \theta} \right| \right. \quad (\text{A.36})$$

$$= i \cot \theta \quad (\text{A.37})$$

This seems wrong, will find out why hopefully.

A.6 squeezed state and a beam splitter

Input field:

$$|\Psi_{\text{in}}\rangle = |\xi\rangle_a |0\rangle_b |0\rangle_c |0\rangle_d \quad (\text{A.38})$$

$$= \exp\left(\frac{1}{2}\xi^* a^2 - \frac{1}{2}\xi a^{\dagger 2}\right) |\vec{0}\rangle. \quad (\text{A.39})$$

Output field:

$$|\Psi_{\text{out}}\rangle = \exp\left(\frac{1}{2}\xi^*\left(\frac{1}{\sqrt{2}}c + \frac{1}{\sqrt{2}}d\right)^2 - \frac{1}{2}\xi\left(\frac{1}{\sqrt{2}}c + \frac{1}{\sqrt{2}}d\right)^{\dagger 2}\right)|\vec{0}\rangle \quad (\text{A.40})$$

$$= \exp\left(\frac{1}{4}\xi^*c^2 - \frac{1}{4}\xi c^{\dagger 2}\right) \exp\left(\frac{1}{4}\xi^*d^2 - \frac{1}{4}\xi d^{\dagger 2}\right) \quad (\text{A.41})$$

$$\times \exp\left(\frac{1}{2}\xi^*(cd) - \frac{1}{2}\xi(cd)^\dagger\right)|\vec{0}\rangle \quad (\text{A.42})$$

$$= S_c\left(\frac{\xi}{2}\right) S_d\left(\frac{\xi}{2}\right) S_{cd}\left(\frac{\xi}{2}\right)|\vec{0}\rangle \quad (\text{A.43})$$

This is a combination of single and two photon squeezing.

A.7 coherent state, squeezed state and a beam splitter

Input field:

$$|\Psi_{\text{in}}\rangle = |\xi\rangle_a |\beta\rangle_b |0\rangle_c |0\rangle_d. \quad (\text{A.44})$$

Output field:

$$|\Psi_{\text{out}}\rangle = S_c\left(\frac{\xi}{2}\right) S_d\left(\frac{\xi}{2}\right) S_{cd}\left(\frac{\xi}{2}\right) D_c\left(\frac{\beta}{2}\right) D_d\left(-\frac{\beta}{2}\right)|\vec{0}\rangle \quad (\text{A.45})$$

Bibliography

- [1] Sebastian Steinlechner, Bryan W. Barr, Angus S. Bell, Stefan L. Danilishin, Andreas Gläfke, Christian Gräf, Jan-Simon Hennig, E. Alasdair Houston, Sabina H. Huttner, Sean S. Leavey, Daniela Pascucci, Borja Sorazu, Andrew Spencer, Kenneth A. Strain, Jennifer Wright, and Stefan Hild. Local-oscillator noise coupling in balanced homodyne readout for advanced gravitational wave detectors. *Physical Review D*, 92(7), oct 2015.
- [2] Carlton M. Caves and Bonny L. Schumaker. New formalism for two-photon quantum optics. i. quadrature phases and squeezed states. *Physical Review A*, 31(5):3068–3092, may 1985.
- [3] Kiwamu Izumi and Daniel Sigg. Advanced LIGO: length sensing and control in a dual recycled interferometric gravitational wave antenna. *Classical and Quantum Gravity*, 34(1):015001, dec 2016.
- [4] D.E. McClelland, N. Mavalvala, Y. Chen, and R. Schnabel. Advanced interferometry, quantum optics and optomechanics in gravitational wave detectors. *Laser & Photonics Reviews*, 5(5):677–696, mar 2011.

- [5] Haocun Yu, , L. McCuller, M. Tse, N. Kijbunchoo, L. Barsotti, and N. Mavalvala. Quantum correlations between light and the kilogram-mass mirrors of LIGO. *Nature*, 583(7814):43–47, jul 2020.
- [6] The LIGO Scientific Collaboration. A gravitational wave observatory operating beyond the quantum shot-noise limit. *Nature Physics*, 7(12):962–965, sep 2011.
- [7] L. McCuller, C. Whittle, D. Ganapathy, K. Komori, M. Tse, A. Fernandez-Galiana, L. Barsotti, P. Fritschel, M. MacInnis, F. Matichard, K. Mason, N. Mavalvala, R. Mittleman, Haocun Yu, M.E. Zucker, and M. Evans. Frequency-dependent squeezing for advanced LIGO. *Physical Review Letters*, 124(17), apr 2020.
- [8] H. J. Kimble, Yuri Levin, Andrey B. Matsko, Kip S. Thorne, and Sergey P. Vyatchanin. Conversion of conventional gravitational-wave interferometers into quantum nondemolition interferometers by modifying their input and/or output optics. *Physical Review D*, 65(2), dec 2001.
- [9] James Lough, Emil Schreiber, Fabio Bergamin, Hartmut Grote, Moritz Mehmet, Henning Vahlbruch, Christoph Affeldt, Marc Brinkmann, Aparna Bisht, Volker Kringel, Harald Lück, Nikhil Mukund, Séverin Nadji, Borja Sorazu, Kenneth Strain, Michael Weinert, and Karsten Danzmann. First demonstration of 6 db quantum noise reduction in a kilometer scale gravitational wave observatory.
- [10] T. Zhang. *Advanced techniques for future generation gravitational wave detectors*. PhD thesis, University of Glasgow, 2019.

- [11] C. Bond, P. Fulda, A. Freise, D. Brown, K. Kokeyama, L. Carbone, and A. Perreca. Finesse Input Files for the L1 Interferometer. LIGO Document T1300901-v5 <https://dcc.ligo.org/LIGO-T1300901>.
- [12] L. Barsotti, L. McCuller, M. Evans, and P. Fritschel. The A+ design curve. LIGO DCC, April 2018. LIGO Document T1800042-v5 <https://dcc.ligo.org/LIGO-T1800042/public>.
- [13] B. Lantz, R. Mittleman, J. Kissel, J. Warner, and A. Pele. Typical Motion of HAM Suspension Points in O2. LIGO Document T1800066-v2 <https://dcc.ligo.org/T1800066>.
- [14] V. Frolov. DARM Optical Gain and Losses. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42231>.
- [15] L. McCuller. Frequency Dependence of Squeezing Loss and Angle, Power Measurement by Radiation Pressure. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=52718>.
- [16] L. McCuller. Frequency Dependent Squeezing effects in the LIGO Interferometers. LIGO Document G2000863-v2 <https://dcc.ligo.org/LIGO-G2000863>.
- [17] V. Frolov. Arm Cavity Power, Power Recycling Gain, IFO Reflectivity and Arm Cavity Losses. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42639>.
- [18] J. Richardson, M. Tse, and C. Austin. Measurement of SRC Gouy Phase. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=39779>.

- [19] C. Blair. OMC Scans. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=37742>.
- [20] C. Blair and V. Frolov. 40W OMC Scans Method and Preliminary Data. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=45830>.
- [21] M. Kasprzack, A. Effler, A. Mullavey, and M. Tse. HAM6 Work - M12 Move. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=37291>.
- [22] A. Effler and M. Kasprzack. LLO Interferometer Output Beam Study. LIGO Document T1800089-v2 <https://dcc.ligo.org/LIGO-T1800089>.
- [23] C. Blair, V. Frolov, and A. Effler. SRM CO2 Chop. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=38689>.
- [24] C. Blair. Annular CO2 1W at Various IFO Power. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=43065>.
- [25] C. Blair, V. Frolov, and A. Effler. OMC Mode Matching in Single Bounce. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=38698>.
- [26] A. Effler, V. Frolov, and C. Blair. OMC Scans of Cooling Interferometer. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=41428>.
- [27] A. Green, A. Effler, P. Dupej, V. Frolov, P. Fulda, and H. Yu. Finesse Squeezing Mode Matching Study for LLO. LIGO Document T1900159-v1 <https://dcc.ligo.org/LIGO-T1900159>.
- [28] A. Effler, P. Dupej, H. Yu, A. Green, and C. Blair. SQZ SRM Bounce 98% Mode Matching. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42187>.

- [29] Daniel David Brown and Andreas Freise. Pykat, July 2017. <http://www.gwoptics.org/pykat>.
- [30] C. Blair. PRC Gouy phase measurement. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=49409>.
- [31] A. A. Muzammil, C. Mueller, and G. Mueller. Optical Layout and Parameters for the Advanced LIGO Cavities. LIGO Document T0900043-v11 <https://dcc.ligo.org/LIGO-T0900043/public>.
- [32] LIGO Optics Database. <https://galaxy.ligo.caltech.edu/optics/>.
- [33] R. Martin and G. Billingsley. Measurement Uncertainties in Determining the Radius of Curvature for SR2-04. LIGO Document T1300653-v1 <https://dcc.ligo.org/LIGO-T1300653>.
- [34] G. Billingsley. SR301 Figure Measurement. LIGO Document E1101231-v1 <https://dcc.ligo.org/E1101231-v1>.
- [35] P. Fritschel and E. Gustafson. Advanced LIGO Optical Layout. LIGO Document D0902838-v5 <https://dcc.ligo.org/LIGO-D0902838>.
- [36] M. Smith and D. Coyne. Advanced LIGO L1 Optical Layout, ZEMAX. LIGO Document D0902216-v8 <https://dcc.ligo.org/LIGO-D0902216>.
- [37] Antonio Perreca, Aidan F. Brooks, Jonathan W. Richardson, Daniel Töyrä, and Rory Smith. Analysis and visualization of the output mode-matching requirements for squeezing in advanced LIGO and future gravitational wave detectors. *Physical Review D*, 101(10), may 2020.